

Cerebras

AI Infrastructure Architecture Evolution

2015-2026

Silicon generations · Memory and interconnects
Conference disclosures · Systems and software

A generation-by-generation synthesis of conference disclosures and primary technical sources.
Evidence cutoff: 28 September 2026.
Complete English edition, paired with a full Simplified Chinese report.

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01 Scope and architectural synthesis

This report covers Cerebras from its 2016 founding through 28 September 2026, within the series' 2015-2026 window. It follows WSE1/2/3/3 Turbo, CS1/2/3/4, MemoryX, SwarmX, wafer I/O, software and the disclosed CS5/CS6 roadmap. It distinguishes wafer silicon from complete systems and deployment services. There is no separate Cerebras server-CPU or merchant DPU line in the documented portfolio.

Cerebras changes the physical boundary of a computer: hundreds of thousands of small cores and their local SRAM communicate on a continuous wafer-scale mesh. This reduces many package crossings, but replaces commodity packaging with demanding yield repair, power delivery, cooling and mapping requirements. Its evolution moves from making wafer-scale computation practical to separating model storage from execution, and then to organizing multiple wafers for fast inference.

02 WSE1 and CS1: making wafer scale work

The 2019 Hot Chips disclosure introduced a 46,225 mm², 16 nm processor with 1.2 trillion transistors, 400,000 cores and 18 GB distributed SRAM. Local memory and a two-dimensional mesh replace the conventional pattern of repeatedly reaching off-chip for operands. The 9 PB/s SRAM figure is the sum of local memory interfaces; the 100 Pb/s fabric figure is an aggregate of mesh links. Neither is a global arbitrary-access bandwidth available to one core.

Architectural interpretation: a wafer-scale device cannot require a defect-free wafer. Redundancy and routing repair are part of the architecture, alongside physical continuity across reticle boundaries. This makes yield a system-design problem rather than merely a transistor-density metric. CS1 adds the power and cooling machinery needed to expose the wafer as usable computing infrastructure.

03 WSE2 and CS2: local dataflow and Weight Streaming

WSE2 moves to 7 nm with 850,000 cores and 40 GB SRAM. The 2022 Hot Chips architecture deep dive describes independently programmed cores with 48 KB local memory, a small software-managed accumulator cache, tensor address generation and data-triggered execution. The mesh extends across stitched reticle regions. Source-synchronous parallel links and repair state machines preserve the logical fabric despite manufacturing defects.

Sparsity is supported through dataflow: zero elements can be filtered before sending, so receiving cores do not execute the corresponding work. This is distinct from advertising a fixed two-times structured-sparse peak. Benefits depend on actual sparsity, mapping and communication balance. The local memories also hold code and working data, so aggregate SRAM capacity cannot be assigned entirely to model weights.

Weight Streaming separates parameter storage from wafer execution. MemoryX stores and streams weights; SwarmX broadcasts weights and reduces gradients across CS systems. Each wafer executes the mapped work while external storage supports models larger than its SRAM. This is not an enlargement of on-wafer SRAM and does not create cache-coherent shared memory. The operator-facing simplification of distributed training still depends on a compiler and fabric that move the right data at the right time.

04 WSE3 and CS3: more compute, similar memory capacity

WSE3 uses 5 nm, four trillion transistors, 900,000 cores and 44 GB SRAM. The advertised 125 PFLOP/s AI compute rate needs a named numerical and sparsity basis before it can enter a dense BF16 comparison. Core count grows about 5.9% over WSE2, while SRAM capacity grows 10%. The generation's performance improvement therefore cannot be explained simply as twice as many cores. ISA, datapath and implementation changes matter.

The 2024 launch describes external memory configurations up to 1.2 PB and clusters up to 2,048 CS3 systems. Those are system/storage scale claims, not SRAM capacity per wafer. Hot Chips 2024 explains training through Weight Streaming and inference through model placement on the wafer fabric. For latency-sensitive decode, keeping weights in SRAM avoids repeated off-chip weight reads, but capacity determines how many wafers and pipeline stages a model needs.

05 CS4 and WSE3 Turbo: a system-generation change

CS4, announced in August 2026, is a three-wafer rack-scale system using WSE3 Turbo, not a product called WSE4. Each wafer retains four trillion transistors, 900,000 cores and 44 GB SRAM. Higher operating rates accompany new power delivery, modular cooling and wafer I/O in the Nexus platform. The published system totals are 132 GB SRAM, 129.6 PB/s SRAM bandwidth and 7.2 Tb/s off-wafer I/O. First shipments were targeted for that quarter; the announcement alone does not establish completed general availability.

The implementation emphasis shifts toward reducing the final power-delivery distance and making compute, power and I/O modules independently serviceable. Moving power conversion close to the wafer reduces resistive loss; it does not remove the need to dissipate the delivered power. The platform's usefulness must be assessed with facility power, coolant requirements, maintenance isolation and failure recovery, not solely wafer arithmetic.

The programmable Wafer I/O Module supports RoCEv2 Ethernet and Direct Wafer Links. The latter connects wafers without an intervening switch, with a vendor-reported latency as low as two microseconds. Per-wafer off-wafer bandwidth is 2.4 Tb/s, or 300 GB/s as a nominal unit conversion; the announcement does not unambiguously specify directional aggregation. This external path remains radically smaller than the sum of local SRAM interfaces, making placement and pipelining essential.

06 Silicon and system specification tables

Wafer specifications, per wafer · 1/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Architecture	GA / milestone	Status	Process	Dies / packaging
WSE1 / CS1	Wafer-scale dataflow	2019	Historical shipping	16 nm	1 wafer / system
WSE2 / CS2	Wafer-scale dataflow	2021	Historical shipping	7 nm	1 wafer / system
WSE3 / CS3	Wafer-scale dataflow	2024	Shipping / available	5 nm	1 wafer / system
WSE3 Turbo / CS4	Wafer-scale dataflow	2026 announcement	Announced; GA n/d	5 nm	3 wafers / CS4; numbers per wafer

Wafer specifications, per wafer · 2/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Compute units	Dense BF16	Dense FP8	Dense FP6 / FP4	FP64 vector / matrix
WSE1 / CS1	400000 cores	n/d	n/d	n/d	n/d
WSE2 / CS2	850000 cores	n/d	n/d	n/d	n/d
WSE3 / CS3	900000 cores	n/d	n/d	n/d	n/d
WSE3 Turbo / CS4	900000 cores	n/d	n/d	n/d	n/d

Wafer specifications, per wafer - 3/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Memory: type; stacks; GB; TB/s	LLC / SRAM
WSE1 / CS1	No attached HBM; external MemoryX separate	18 GB SRAM; 9 PB/s aggregate
WSE2 / CS2	No attached HBM; external MemoryX separate	40 GB SRAM; 20 PB/s aggregate
WSE3 / CS3	No attached HBM; external MemoryX separate	44 GB SRAM; 21.6 PB/s aggregate
WSE3 Turbo / CS4	No attached HBM; external MemoryX separate	44 GB SRAM; 43.2 PB/s aggregate

Wafer specifications, per wafer - 4/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Scale-up: links; GB/s per direction; domain	Host link	Power / cooling	Form factor
WSE1 / CS1	2D mesh; per-link n/d; one wafer	System I/O; not host PCIe card	n/d; liquid	Wafer-scale system
WSE2 / CS2	2D mesh; per-link n/d; one wafer	System I/O; not host PCIe card	n/d; liquid	Wafer-scale system
WSE3 / CS3	2D mesh; per-link n/d; one wafer	System I/O; not host PCIe card	n/d; liquid	Wafer-scale system
WSE3 Turbo / CS4	2D mesh; per-link n/d; one wafer	System I/O; not host PCIe card	n/d; liquid	Wafer-scale system

The 2024 slides round WSE3 memory bandwidth to 21 PB/s; the 2026 system comparison uses 21.6 PB/s. This report uses the latter for normalized calculations. An older developer page states a 20% core-count increase, which conflicts with the explicitly published 850,000 and 900,000 counts; the counts are retained. Published fabric figures such as 214 or 220 Pb/s are aggregate link metrics and must not be treated as bisection bandwidth.

Host CPU boundary · 1/3

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Generation / reference SKU	Codename	GA	Status	Core μarch	Cores / threads
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External preprocessing / management CPUs	n/a	n/a	Supplier products	n/a	n/a
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Host CPU boundary · 2/3

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Generation / reference SKU	Process per die	Dies / packaging	L2 per core	L3 / domain
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External preprocessing / management CPUs	n/a	n/a	n/a	n/a
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Host CPU boundary · 3/3

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Generation / reference SKU	Memory: channels; rate; GB/s	Sockets / links	PCIe / CXL	TDP	Vector / matrix ISA
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External preprocessing / management CPUs	n/a	n/a	n/a	n/a	n/a
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System generations · 1/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Platform	Year / milestone	CPU / accelerator count	Scale-up domain
CS1	2019	1 WSE1	1 wafer; cluster topology n/d
CS2 + MemoryX/Swar mX	2021	1 WSE2 per CS2	Weight Streaming cluster
CS3	2024	1 WSE3; up to 2048 systems	SwarmX / workload placement
CS4 / Nexus	2026 announcement	3 WSE3T	Direct Wafer Links / Ethernet
CS5 / CS6	2027 target / later	n/d	n/d

System generations · 2/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Platform	Scale-out per accelerator	Rack power	Cooling
CS1	n/d	n/d	Liquid
CS2 + MemoryX/Swar mX	n/d	n/d	Liquid
CS3	1.2 Tb/s system I/O; direction n/d	n/d	Liquid
CS4 / Nexus	2.4 Tb/s per wafer; direction n/d	n/d	Modular liquid cooling
CS5 / CS6	n/d	n/d	n/d

07 MemoryX, SwarmX and networking boundaries

Communication products and functions · 1/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	GA / milestone	Status	Class
SwarmX	2021	Shipping / available	Broadcast/reduce fabric
Wafer I/O Module	2026	Announced; GA n/d	Programmable wafer I/O
Merchant DPU / host NIC	n/a	n/a	n/a

Communication products and functions · 2/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Ports × speed	SerDes	Host interface	Programmable engine
SwarmX	n/d	n/d	System Ethernet	Collective acceleration
Wafer I/O Module	2.4 Tb/s/wafer aggregate; direction n/d	n/d	Wafer system interface	Programmable I/O
Merchant DPU / host NIC	n/a	n/a	n/a	n/a

Communication products and functions · 3/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Embedded CPU	On-card memory	Transport / RDMA	Congestion control
SwarmX	n/d	n/d	Weight/gradient streaming	n/d
Wafer I/O Module	n/d	n/d	RoCEv2 / Direct Wafer Links	n/d
Merchant DPU / host NIC	n/a	n/a	n/a	n/a

Communication products and functions · 4/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Key offloads	Power
SwarmX	Broadcast / reduction	n/d
Wafer I/O Module	Wafer-to-wafer data movement	n/d
Merchant DPU / host NIC	n/a	n/a

Interconnect hierarchy · 1/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Interconnect	Year / generation	Lane rate	Lanes / link	GB/s per link per direction
On-wafer mesh	WSE1-3T	Generation-specific	32-bit interfaces in HC2022	n/d
SwarmX	2021 onward	n/d	n/d	n/d
Direct Wafer Links	2026	n/d	n/d	n/d

Interconnect hierarchy · 2/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Interconnect	Links / device	Topology	Domain limit	Coherence / semantics
On-wafer mesh	4 neighbors/core	2D mesh	Wafer	Explicit local-memory messages
SwarmX	n/d	Broadcast / reduction tree	System configuration	Weight streaming; not coherent
Direct Wafer Links	n/d	Switchless wafer links	n/d	Explicit communication

Four bandwidth levels must remain distinct: a core’s SRAM port, the sum of SRAM ports across a wafer, the wafer mesh under a particular traffic pattern, and off-wafer I/O. Adding them or comparing their totals directly with GPU rack bisection obscures bottlenecks. A model that communicates mainly between neighbors can exploit the mesh differently from all-to-all expert routing or a global transpose.

08 Programming, deployment and heterogeneous inference

The software has two useful levels: framework/model compilation for AI users and low-level spatial/dataflow programming for kernels and scientific workloads. The compiler must allocate local storage, place work and route communication while respecting defects and resource limits. Weight Streaming hides much distributed-training complexity from the user, but does not make these physical constraints disappear. SC stencil research and wafer-scale FFT work demonstrate alternative mappings, not universal speedups for every scientific application.

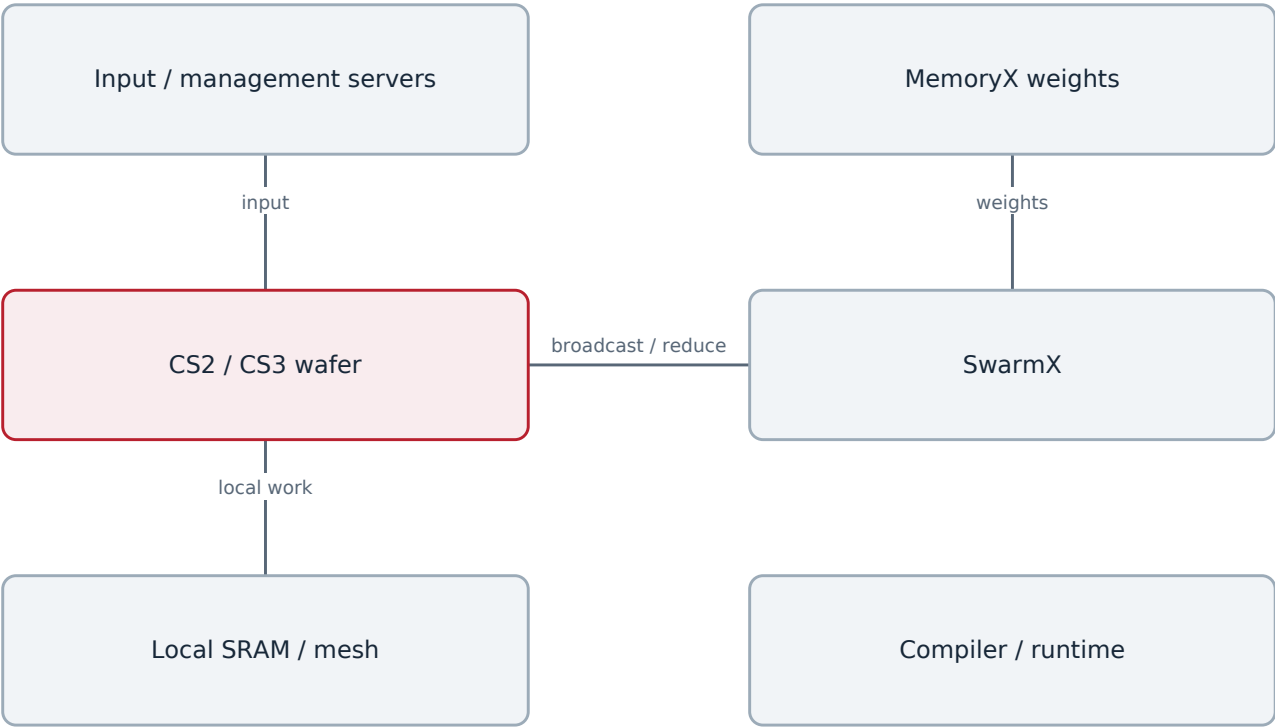
The March 2026 AWS collaboration describes CS3 deployment and a planned disaggregated inference path: Trainium performs prefill and transfers KV state over EFA to Cerebras for decode. This makes the phase boundary a system interface. The benefit depends on prompt length, output length, batching, transfer time and queueing. The announcement is not used as proof of public general availability for every named model.

Software and control layers

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Layer	Component	Function	Constraint
Model	Framework / model compiler	Compile training and inference	Operator and numerical support
Kernel	CSL / dataflow mapping	Spatial compute and communication	Local memory and mesh routing
Training	MemoryX / SwarmX control	Weights and gradient reduction	Streaming schedule and bandwidth
Operations	Management / preprocessing servers	Input and resource scheduling	External hosts and services

2021-2024 training stack



MemoryX capacity is external to the wafer SRAM.

2026 inference integration



Functional options from separate announcements; not a single confirmed shipping configuration.

09 Derived balance and future architectural direction

Normalized metrics

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Metric	Calculation	Meaning
WSE2→WSE3 core growth	$900000/850000 = 1.059$	5.9%, not 20%
WSE3→CS4 SRAM	$3 \times 44 = 132 \text{ GB}$	System aggregation; not one wafer
WSE3T SRAM/external I/O	$43.2 \text{ PB/s} / 0.0003 \text{ PB/s} = 144000$	Scale contrast; direction and topology differ
CS4 I/O unit conversion	$7.2 \text{ Tb/s} / 8 = 900 \text{ GB/s}$	Nominal aggregate; direction n/d
44 GB / BF16 weight	$44e9/2 = 22 \text{ billion parameters}$	Capacity upper bound before code/KV/workspace
Dense HBM/FLOP	n/a	SRAM architecture; no HBM tier
SRAM/FLOP; capacity/FLOP; FLOP/W	n/d	Matched dense format and power basis absent
Host DDR/core; L3/core	n/a	No proprietary host CPU

At Hot Chips 2026, Cerebras described CS5 for 2027 using a next-generation WSE, and CS6 as a future combination of wafer-scale SRAM/compute with stacked DRAM. These are roadmap disclosures. The architectural importance of CS6 is a new memory tier: capacity could grow without proportionally multiplying wafers, but locality, stacking yield, thermal resistance and the SRAM/DRAM interface become central. No undisclosed capacity, process or measured speed is assigned to it.

10 Conference index and naming crosswalk

Disclosure map · 1/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Venue	Year	Paper / talk / disclosure
Hot Chips	2019	Wafer Scale Engine
Hot Chips	2021	Extreme-scale AI / Weight Streaming
Hot Chips	2022	Architecture deep dive
Hot Chips	2023	Inside the Wafer-Scale Cluster
Hot Chips	2024	Wafer-Scale AI
Hot Chips	2026	Ultrafast frontier inference
ISPD	2020	Wafer-scale placement contest
SC	2020	Fast Stencil-Code Computation

Disclosure map · 2/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Venue	Presenter / authors	Product	Evidence type	Source
Hot Chips	Cerebras / research authors	WSE1	Architecture / implementation / workload	E05
Hot Chips	Cerebras / research authors	WSE2 / CS2	Architecture / implementation / workload	E06 E03
Hot Chips	Cerebras / research authors	Core / SRAM / mesh	Architecture / implementation / workload	C01
Hot Chips	Cerebras / research authors	MemoryX / SwarmX	Architecture / implementation / workload	C06
Hot Chips	Cerebras / research authors	WSE3 / CS3	Architecture / implementation / workload	C02
Hot Chips	Cerebras / research authors	CS4; CS5/6 roadmap	Architecture / implementation / workload	C03
ISPD	Cerebras / research authors	Physical mapping	Architecture / implementation / workload	C05
SC	Cerebras / research authors	CS1 research workload	Architecture / implementation / workload	C07

Hot Chips provides the main product architecture chain; ISPD and SC illuminate mapping and application behavior. This edition does not establish a direct product implementation paper at ISCA, MICRO, HPCA, ASPLOS or ISSCC. IEEE Micro journal articles derived from Hot Chips must not be mislabelled MICRO conference papers. Detailed comparison still needs matched numerical-format peaks, system power and workload-level throughput under a stated service latency.

Names and counting levels

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Name	Meaning	Boundary
WSE	Wafer processor	Silicon generation
CS	Complete compute system	CS4 uses three WSE3T
MemoryX	External model storage/streaming	Not on-wafer SRAM
Swarm / SwarmX	On-wafer fabric / cluster fabric	Different physical boundaries
Nexus	Rack platform	Power, cooling, I/O modularity

Source map

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- 02 WSE1 and CS1: making wafer scale work - C01, E05
- 03 WSE2 and CS2: local dataflow and Weight Streaming - C01, C02, E03, E06, P01
- 04 WSE3 and CS3: more compute, similar memory capacity - C02, E01, P01
- 05 CS4 and WSE3 Turbo: a system-generation change - C03, E02, P02
- 06 Silicon and system specification tables - C02, C03, E01, E02, E03, E05, E06, P01
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- 08 Programming, deployment and heterogeneous inference - C01, C04, C07, E01, E03, E04, P01
- 09 Derived balance and future architectural direction - C02, C03, E01, E02, E06, P01
- 10 Conference index and naming crosswalk - C01, C02, C03, C05, C06, C07, E01, E02, E03, E05, E06, P01

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